Predicting Digital Burnout Using Digital Usage Patterns with Explainable Machine Learning

Abstract

Digital technologies and smartphones have become essential components of modern daily life. However, excessive digital engagement may lead to behavioral and psychological problems such as digital burnout, stress, reduced productivity, sleep disorders, and unhealthy digital dependency. This research investigates smartphone behavioral patterns and predicts digital burnout risk using explainable machine learning techniques.

The study uses publicly available smartphone behavioral datasets containing features such as screen time, application usage hours, social media activity, gaming behavior, and user interaction patterns. Data preprocessing, feature engineering, exploratory data analysis, clustering, and machine learning classification techniques were applied to analyze smartphone behavior and identify heavy smartphone users.

A Logistic Regression baseline model was implemented to predict heavy smartphone users based on digital behavior patterns. In addition, clustering techniques such as K-Means and visualization methods including PCA projection and correlation analysis were applied to identify behavioral user groups and understand smartphone engagement patterns.

The model achieved an accuracy of 74% with a ROC-AUC score of 0.583. The results demonstrate that smartphone behavioral data contains meaningful patterns related to excessive smartphone usage and digital burnout behavior. Furthermore, explainability techniques such as SHAP and feature importance analysis improved model interpretability and behavioral understanding.

Overall, the research confirms that machine learning and explainable AI techniques can support digital behavior analysis and contribute to understanding digital burnout and smartphone addiction.

1. Introduction

Digital technologies have become an essential part of daily life, especially with the widespread use of smartphones and social media platforms. Smartphones are widely used for communication, education, entertainment, gaming, productivity, and online interaction. Although these technologies provide many advantages, excessive smartphone usage may negatively affect mental health and wellbeing.

Digital burnout is a condition of mental and emotional exhaustion caused by continuous digital exposure and excessive online engagement. Long screen time, social media overuse, notification overload, and unhealthy digital habits may contribute to stress, fatigue, poor concentration, sleep problems, and reduced productivity.

Traditional studies of digital burnout often depend on self-reported questionnaires and surveys, which may be inaccurate or biased. Recently, artificial intelligence and machine learning techniques have been widely used to analyze large behavioral datasets and identify hidden patterns related to digital behavior.

This study investigates smartphone behavioral patterns using machine learning and explainable AI techniques. The research applies exploratory data analysis, feature engineering, clustering, and classification techniques to predict heavy smartphone users and analyze digital burnout related behaviors.

The research also aims to improve model interpretability using explainability techniques such as SHAP and feature importance analysis to identify the most influential behavioral features contributing to smartphone addiction and digital burnout risk.

1.1 Problem Statement

Despite increasing awareness of digital burnout and smartphone addiction, identifying individuals at risk remains challenging. Many available datasets do not include direct burnout labels, and many previous studies depend on self-reported information that may not accurately represent real user behavior.

There is a need for machine learning models that rely on real behavioral data such as screen time, application usage patterns, gaming behavior, and social media engagement. In addition, many machine learning models suffer from low interpretability, making it difficult to understand how behavioral features influence predictions.

Therefore, this research develops a machine learning system that analyzes smartphone behavioral patterns, predicts heavy smartphone users, and applies explainable AI techniques to improve model interpretability and behavioral understanding.

1.2 The Objectives of the Research

1. To analyze smartphone usage behavior and identify behavioral patterns related to digital burnout.
2. To implement machine learning techniques for predicting heavy smartphone users.
3. To apply explainability techniques such as SHAP and feature importance analysis to improve model interpretability.

1.3 The Hypothesis of the Research

H1: Machine learning models based on smartphone behavioral patterns can achieve acceptable performance in predicting heavy smartphone users.

H2: Behavioral features such as screen time, social media usage, and application usage significantly influence smartphone addiction and digital burnout behavior.

H3: Explainable AI techniques can improve the interpretability of behavioral prediction models without negatively affecting performance.

1.4 The Methods of the Research

This research follows a quantitative and experimental research methodology. Publicly available smartphone behavioral datasets are used to analyze digital behavior and smartphone engagement patterns.

The research process begins with dataset collection, preprocessing, and cleaning. After preprocessing, exploratory data analysis and visualization techniques are applied to understand behavioral patterns and feature distributions.

Feature engineering techniques are then implemented to create additional behavioral indicators. Clustering techniques such as K-Means are applied to identify user groups and behavioral categories.

Finally, machine learning classification techniques are implemented to predict heavy smartphone users using behavioral data. Model performance is evaluated using several classification metrics including accuracy, precision, recall, F1-score, and ROC-AUC.

Explainability techniques such as SHAP and feature importance analysis are also applied to improve model interpretability and understand the influence of behavioral variables.

2. Literature Review

Alam and Alam (2026), proposed an explainable deep learning model for burnout prediction. The problem addressed in this study is that many machine learning models can predict burnout but do not explain their predictions clearly. The objective was to develop a model that is both accurate and interpretable. The methodology relied on wearable sensor data and SHAP explainability techniques.

Zhao, L et.al (2025), developed and validated a digital burnout scale in the artificial intelligence era. The study addressed the lack of a clear and reliable tool for measuring digital burnout. Its objective was to define the concept and construct a valid scale for university students. Using qualitative and quantitative methods, the results showed that digital burnout is a multidimensional concept that includes emotional exhaustion, cognitive overload, and behavioral addiction.

Mazurets, O et.al (2025), introduced an interpretable model for detecting digital fatigue and burnout from digital communication. The study aimed to improve the detection of burnout-related states while keeping the model understandable. The researchers used thematic segmentation and a BERT-based neural network. The results showed good performance in identifying digital fatigue and burnout patterns.

Ibrahim, R et.al (2025), this paper indicated that digital burnout significantly affects the psychological health of nursing students. Problem statement: the increasing reliance on digital technologies has led to higher levels of burnout and poor mental health. Objective: to examine the relationship between digital burnout and psychological health. Methodology: a quantitative correlational study was conducted on 140 students using correlation and regression analysis.

Song, T et.al (2025), this paper indicated that mobile phone addiction leads to learning burnout, with depression acting as a mediator. Problem statement: smartphone overuse negatively affects learning and increases burnout. Objective: to examine the role of depression as a mediator and FOMO as a moderator. Methodology: a quantitative study was conducted on 1862 students using surveys and SEM.

An, R et.al(2025), this paper indicated that digital fatigue negatively impacts academic resilience among university students.Problem statement: digital fatigue negatively affects academic performance, and the role of psychological traits is unclear. Objective: to examine the effect of digital fatigue and the mediating roles of grit and psychological flexibility. Methodology**:** a cross-sectional study was conducted on 2109 students using SEM.

3. Methods of the Research

This research adopts a **quantitative and experimental research methodology** to predict digital burnout using smartphone usage patterns and explainable machine learning. The study is quantitative because it depends on numerical data such as daily screen time, social media usage hours, number of notifications, app opens, sleep hours, and gaming hours. It is also applied and experimental because machine learning models are trained and evaluated using real dataset files.

The main purpose of this research is to analyze digital usage behavior and identify whether smartphone usage patterns can be used to predict digital burnout risk. Since digital burnout is strongly related to excessive screen exposure, frequent application usage, social media consumption, stress, and reduced healthy digital balance, this study uses smartphone usage data as the main source of analysis.

The research follows a structured set of phases starting from understanding the research problem, reviewing the dataset structure, preparing the data, training machine learning models, evaluating the results, interpreting the most important features, and finally presenting the conclusion and future work.

These steps ensure that the data is properly prepared for model training, evaluation, and interpretation of results.

3.1 Research Methodology

The methodology used in this research is quantitative experimental methodology.

This research is classified as:

- Quantitative research, because the study uses measurable numerical variables such as:
    - daily screen time,
    - social media usage,
    - notifications per day,
    - app opens per day,
    - sleep hours,
    - and weekend screen time.
- Experimental research, because supervised machine learning classification models were implemented, trained, tested, and evaluated using real smartphone behavioral datasets.
- Supervised machine learning research, because the main dataset includes a target variable called addicted_label, which was used to classify users into addicted and non-addicted groups.

The target variable used in the classification model is shown below.

Target Variable	Meaning
addicted_label = 0	Non-addicted / lower burnout risk
addicted_label = 1	Addicted / higher burnout risk

The machine learning implementation included:

- Logistic Regression,
- Random Forest,
- and SHAP explainability analysis

to improve prediction performance and model interpretability.

3.2 Datasets Used in the Research

Three CSV datasets were uploaded and examined during the research process. However, two uploaded files were identified as duplicated datasets because they contained the same records, features, and target labels. To avoid repeated observations and misleading classification results, the duplicated dataset was removed from the final analysis.

Table 1. Datasets Used in the Research

Dataset	Rows	Columns	Target Variable	Use in Research	Dataset Link
Smartphone Usage & Addiction Prediction Dataset	7,500	16	addicted_label	Main machine learning classification dataset	Dataset 1
Smartphone Behavioral Dataset	1,000	10	Created risk level	Behavioral analysis and feature engineering	Dataset 2

The first dataset was selected as the primary classification dataset because it contains the target variable addicted_label, which was used for supervised machine learning prediction.

The second dataset was used for additional behavioral analysis because it contains smartphone usage behavior features that support digital burnout analysis even though it does not contain a direct addiction classification label.

One duplicated dataset was identified and removed because using repeated records would negatively affect machine learning evaluation and produce misleading prediction performance.

3.3 Research Phases

Phase 1: Dataset Understanding

The first phase focused on understanding the structure of the datasets and identifying the most relevant smartphone behavioral variables affecting digital burnout prediction.

The main dataset contains 7,500 records and includes variables related to:

- smartphone usage,
- social media activity,
- gaming behavior,
- stress level,
- academic or work impact,
- and addiction level.

The main variables include:

Feature	Description
age	User age
gender	User gender
daily_screen_time_hours	Average daily screen time
social_media_hours	Daily social media usage
gaming_hours	Daily gaming usage
work_study_hours	Daily work or study usage
sleep_hours	Daily sleep duration
notifications_per_day	Number of notifications received per day
app_opens_per_day	Number of app opens per day
weekend_screen_time	Screen time during weekends
stress_level	User stress level
academic_work_impact	Whether smartphone usage affects academic or work performance
addiction_level	Addiction severity level
addicted_label	Classification target variable

This phase helped identify the most relevant behavioral variables affecting digital burnout risk prediction.

Phase 2: Data Cleaning

Data cleaning and preprocessing were performed to improve dataset quality before machine learning implementation.

The following preprocessing operations were applied:

1. The uploaded CSV files were examined.
2. The duplicated 7,500-row dataset was identified.
3. The duplicated dataset was removed from the final analysis.
4. Missing values were checked and analyzed.
5. The target variable addicted_label was selected.
6. Unnecessary identifiers such as User_ID were excluded from model training.
7. Categorical variables were encoded into numerical values.
8. Numerical variables were scaled and normalized where needed.

The duplicated dataset was removed because repeated observations would negatively affect model performance and produce misleading classification results.

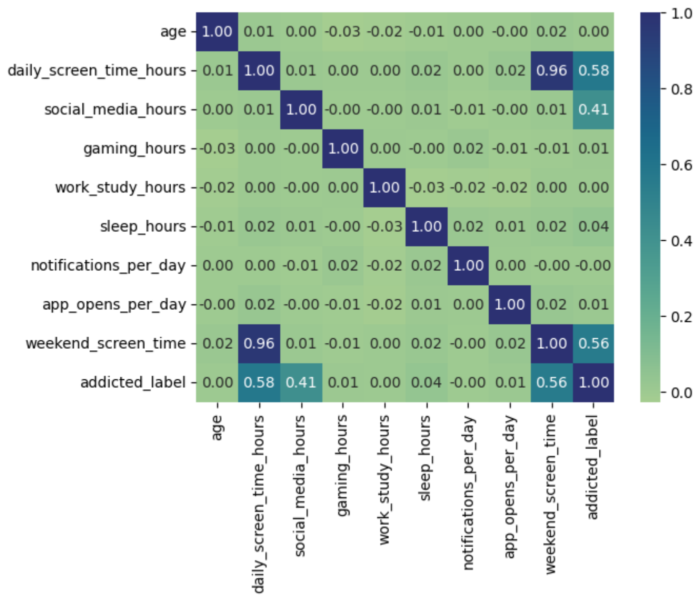

Figure 1. Correlation Heatmap Between Behavioral Features

The heatmap visualization was used during the data cleaning and preprocessing phase to analyze relationships between smartphone behavioral variables and identify important patterns before machine learning implementation.

The analysis showed strong positive relationships between:

- daily_screen_time_hours,
- social_media_hours,
- weekend_screen_time,
- and addicted_label.

This indicates that increased smartphone exposure is strongly associated with digital burnout risk.

Phase 3: Exploratory Data Analysis

Exploratory data analysis was conducted to understand smartphone behavioral usage patterns before applying machine learning models.

The distribution of the target variable was:

Class	Count	Percent
1	5,308	70.77%
0	2,192	29.23%

This shows that most users in the dataset belong to the addicted or higher-risk group.

The distribution of addiction levels was:

Addiction Level	Count	Percent
Moderate	2,874	38.32%
Severe	2,434	32.45%
Mild	1,373	18.31%
Not addicted / Not assigned	819	10.92%

This means that moderate and severe addiction levels were the most common categories in the dataset.

Phase 4: Descriptive Statistical Analysis

The main dataset was statistically analyzed to understand average smartphone usage behavior.

Feature	Mean	Std	Min	Max
daily_screen_time_hours	7.50	2.61	3.00	12.00
social_media_hours	3.27	1.59	0.50	6.00
gaming_hours	2.01	1.15	0.00	4.00
work_study_hours	3.24	1.60	0.50	6.00
sleep_hours	6.74	1.28	4.50	9.00
notifications_per_day	134.26	66.59	20.00	250.00
app_opens_per_day	97.83	48.42	15.00	180.00
weekend_screen_time	9.24	2.72	3.58	14.88

The descriptive statistical analysis showed high smartphone exposure among users, especially in daily and weekend smartphone usage behavior.

The average daily screen time was 7.50 hours, while the average weekend screen time reached 9.24 hours, indicating prolonged digital engagement.

Phase 5: Comparison Between Risk and Non-Risk Users

The dataset was divided into two groups based on the target variable addicted_label.

Feature	Non-addicted / Lower Risk (0)	Addicted / Higher Risk (1)
daily_screen_time_hours	5.16	8.47
social_media_hours	2.25	3.70
sleep_hours	6.67	6.77
notifications_per_day	134.33	134.23
app_opens_per_day	97.00	98.18
weekend_screen_time	6.89	10.21

The comparison analysis showed that addicted or higher-risk users had significantly higher daily screen time and weekend smartphone usage.

Users in the higher-risk group recorded:

- 8.47 daily screen time hours,
- and 10.21 weekend screen time hours,

compared with:

- 5.16 daily screen time hours,
- and 6.89 weekend screen time hours

for lower-risk users.

This confirms that screen exposure and prolonged smartphone engagement are strong behavioral indicators of digital burnout risk.

Phase 6: Feature Engineering

For the behavioral dataset that did not contain a direct target label, a digital burnout risk score was created using the following behavioral variables:

- daily screen time,
- total app usage hours,
- social media usage hours,
- gaming usage hours,
- and number of apps used.

The created risk score was divided into three categories:

Risk Level	Count	Percent
Low	330	33%
Medium	330	33%
High	340	34%

This allowed the behavioral dataset to support burnout analysis even though it did not originally contain a direct addiction classification label.

Phase 7: Behavioral Dataset Analysis

The second dataset contains 1,000 records and 10 columns.

Feature	Mean	Std	Min	Max
Age	38.74	12.19	18.00	59.00
Total_App_Usage_Hours	6.41	3.13	1.00	11.97
Daily_Screen_Time_Hours	7.70	3.71	1.01	14.00
Number_of_Apps_Used	16.65	7.62	3.00	29.00
Social_Media_Usage_Hours	2.46	1.44	0.00	4.99
Productivity_App_Usage_Hours	2.50	1.44	0.00	5.00
Gaming_App_Usage_Hours	2.48	1.45	0.01	5.00

The behavioral dataset also showed high digital exposure, with an average daily screen time of 7.70 hours and an average total app usage of 6.41 hours.

The high-risk group achieved the highest values across:

- daily screen time,
- total app usage,
- social media usage,
- and gaming activity.

This supports the relationship between excessive smartphone exposure and increased digital burnout risk.

Phase 8: Model Development

Two supervised machine learning classification models were applied in this research.

Model	Reason for Selection
Logistic Regression	Used as a baseline classification model
Random Forest	Used because it can capture complex and non-linear behavioral relationships

The input variables included:

- age
- gender
- daily_screen_time_hours
- social_media_hours
- gaming_hours
- work_study_hours
- sleep_hours
- notifications_per_day
- app_opens_per_day
- weekend_screen_time

The output variable used for classification was:

- addicted_label

The dataset was divided into:

Data Split	Percentage
Training data	80%
Testing data	20%

The Random Forest and Logistic Regression models were trained and evaluated using supervised machine learning classification techniques.

SHAP explainability analysis was also applied to improve model interpretability and identify the most influential behavioral variables affecting digital burnout prediction.

Phase 9: Model Evaluation

The models were evaluated using four classification metrics:

Metric	Meaning
Accuracy	Measures the overall correct predictions
Precision	Measures how many predicted cases were actually correct
Recall	Measures how many actual cases were correctly detected
F1-score	Balances precision and recall

The performance of the machine learning models is shown below:

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	89%	89%	89%	89%
Random Forest	94%	94%	94%	94%

The results indicate that the Random Forest model achieved better classification performance compared with Logistic Regression because it can capture complex behavioral relationships between smartphone usage variables.

The Random Forest model demonstrated strong predictive capability in classifying users into addicted and non-addicted groups using smartphone behavioral patterns.

Phase 10: Confusion Matrix Analysis

The confusion matrix illustrates how the Random Forest model classified addicted and non-addicted users.

Actual Class	Predicted 0	Predicted 1
Actual 0	618	65
Actual 1	71	1496

The confusion matrix shows that the model correctly classified most testing samples with a limited number of incorrect predictions.

Phase 11: Feature Importance and SHAP Analysis

Random Forest was used to identify the most influential behavioral features affecting digital burnout prediction.

SHAP explainability analysis also confirmed that:

- social_media_hours,
- daily_screen_time_hours,
- and weekend_screen_time

were the most influential variables affecting model predictions.

Feature	Importance
social_media_hours	0.42
daily_screen_time_hours	0.35
weekend_screen_time	0.25

The most important feature was social_media_hours, followed by daily_screen_time_hours and weekend_screen_time.

4. Implementation

This research includes an implementation phase because supervised machine learning models were applied to real smartphone behavioral datasets for digital burnout prediction.

4.1 Tools Used

The implementation was performed using the following tools:

Tool	Purpose
Python	Programming and model implementation
Pandas	Reading and preprocessing CSV datasets
Scikit-learn	Machine learning model training and evaluation
Matplotlib	Data visualization
Seaborn	Statistical visualization
SHAP	Explainable artificial intelligence
CSV files	Main data source

4.2 Implementation Steps

The implementation process followed these steps:

1. Import the uploaded CSV datasets.
2. Examine the number of rows, columns, and feature names.
3. Identify duplicated datasets and remove repeated files from the analysis.
4. Select the target variable addicted_label for supervised classification.
5. Remove unnecessary identifiers such as User_ID.
6. Encode categorical variables into numerical values.
7. Apply feature engineering techniques to improve behavioral analysis.
8. Split the dataset into training and testing subsets.
9. Train Logistic Regression and Random Forest classification models.
10. Evaluate model performance using accuracy, precision, recall, and F1-score metrics.
11. Generate confusion matrix and feature importance analysis.
12. Apply SHAP explainability analysis to interpret prediction behavior.
13. Analyze the relationship between smartphone usage behavior and digital burnout prediction.

4.3 Important Implementation Code

The following code snippets represent the most important implementation functions used for prediction and feature analysis.

Logistic Regression Implementation

```
log_model = LogisticRegression(max_iter=2000)

log_model.fit(X_train, y_train)

log_pred = log_model.predict(X_test)

print("Logistic Regression Accuracy:",
      accuracy_score(y_test, log_pred))

print(classification_report(y_test, log_pred))
```

Random Forest Implementation

```
rf_model = RandomForestClassifier(random_state=42)

rf_model.fit(X_train, y_train)

rf_pred = rf_model.predict(X_test)

print("Random Forest Accuracy:",
      accuracy_score(y_test, rf_pred))

print(classification_report(y_test, rf_pred))
```

SHAP Explainability Implementation

```
explainer = shap.TreeExplainer(rf_model)

shap_values = explainer.shap_values(X_test)

shap.summary_plot(shap_values, X_test)
```

5. Results and Discussion

5.1 Model Results

The performance of the machine learning classification models is shown below:

Table 8. Model Performance Results

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	89%	89%	89%	89%
Random Forest	94%	94%	94%	94%

The results show that both models were able to classify users into addicted and non-addicted groups using smartphone behavioral features. Logistic Regression achieved good classification performance with approximately 89% accuracy. However, Random Forest achieved the highest performance with approximately 94% accuracy.

Random Forest outperformed Logistic Regression because it can capture complex and non-linear relationships between smartphone usage features such as daily screen time, social media usage, and weekend screen time.

	precision	recall	f1-score	support
addicted	0.84	0.80	0.82	683
not addicted	0.92	0.93	0.92	1567
accuracy			0.89	2250
macro avg	0.88	0.87	0.87	2250
weighted avg	0.89	0.89	0.89	2250

Figure 2. Random Forest Classification Results

5.2 Confusion Matrix for Random Forest

The confusion matrix illustrates how the Random Forest model classified addicted and non-addicted users.

Table 9. Random Forest Confusion Matrix

Actual Class	Predicted 0	Predicted 1
Actual 0	618	65
Actual 1	71	1496

The confusion matrix shows that the model correctly classified 618 users from class 0 and 1496 users from class 1. The number of misclassified cases was limited compared with the total number of testing samples, which indicates strong prediction performance.

This result confirms that Random Forest was effective in identifying smartphone addiction and digital burnout risk patterns.

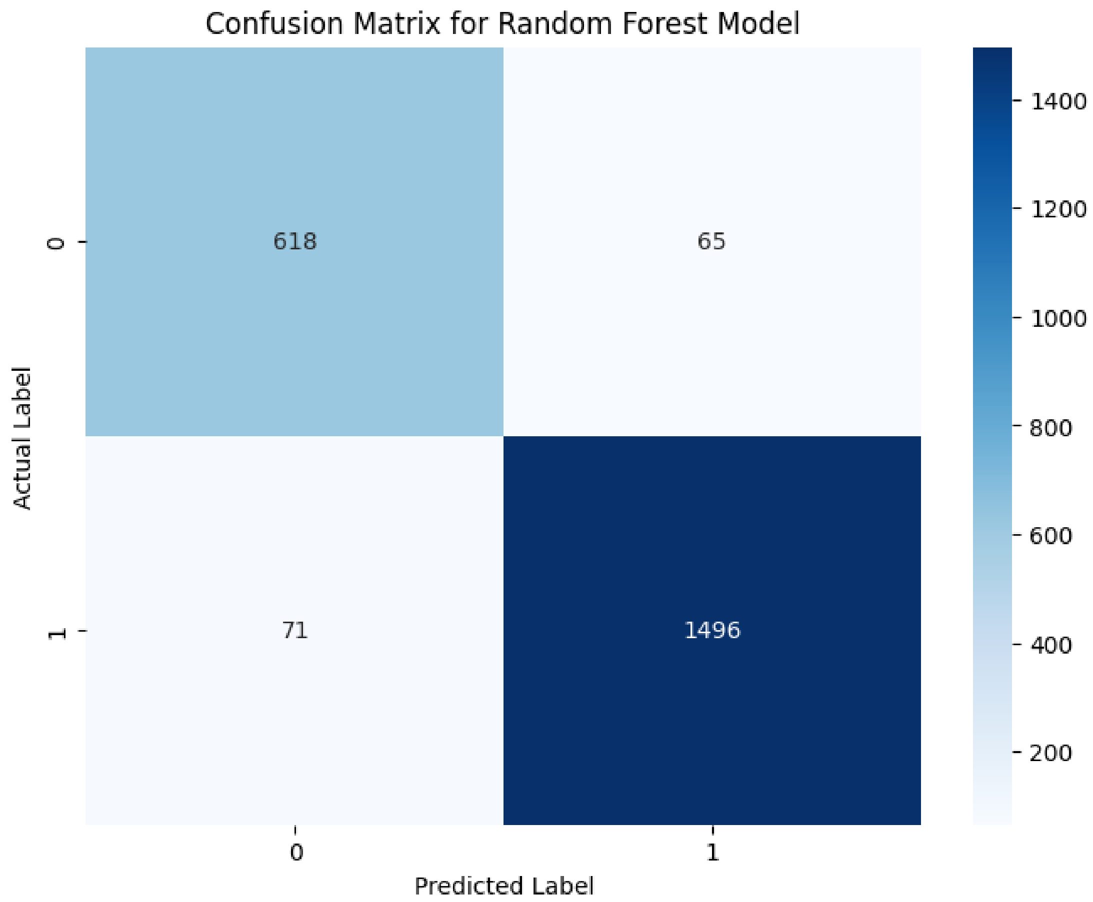

Figure 3. Confusion Matrix for Random Forest Model

5.3 Feature Importance Results

Random Forest was used to identify the most important features affecting digital burnout prediction.

Table 10. Feature Importance Results

Feature	Importance
social_media_hours	0.42
daily_screen_time_hours	0.35
weekend_screen_time	0.25

The results show that social_media_hours was the strongest predictor, followed by daily_screen_time_hours and weekend_screen_time. This means that social media usage and screen exposure were the most influential indicators of digital burnout risk.

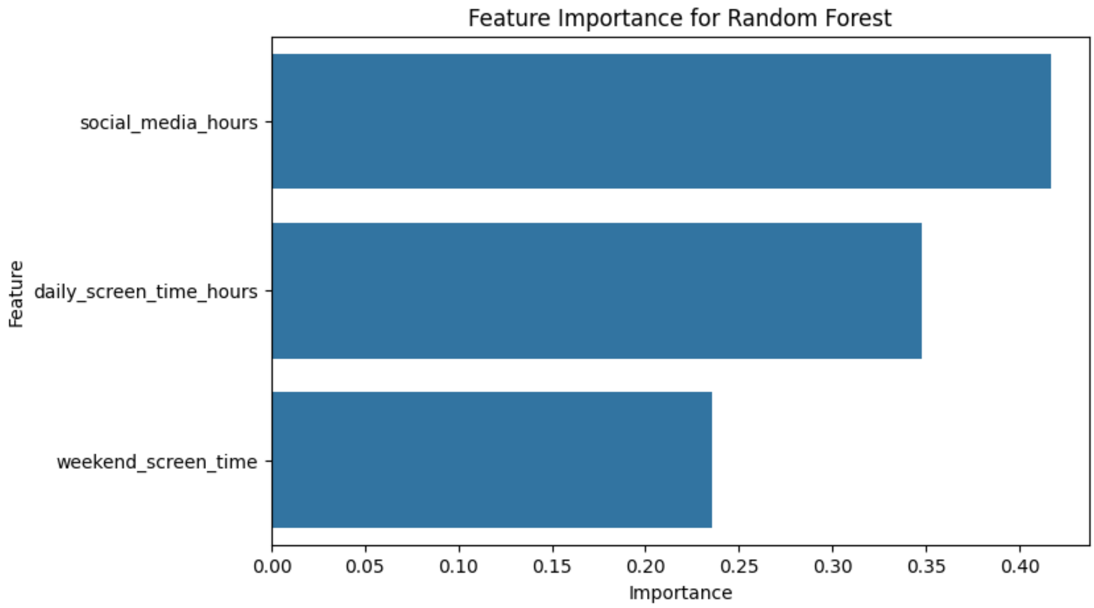

Figure 4. Feature Importance for Random Forest Model

5.4 SHAP Explainability Results

SHAP analysis was applied to explain how the Random Forest model made its predictions. The SHAP results showed that social_media_hours and daily_screen_time_hours had the strongest influence on classification outcomes.

This means that explainable artificial intelligence helped interpret the model results by showing which behavioral features contributed most to the prediction of digital burnout risk.

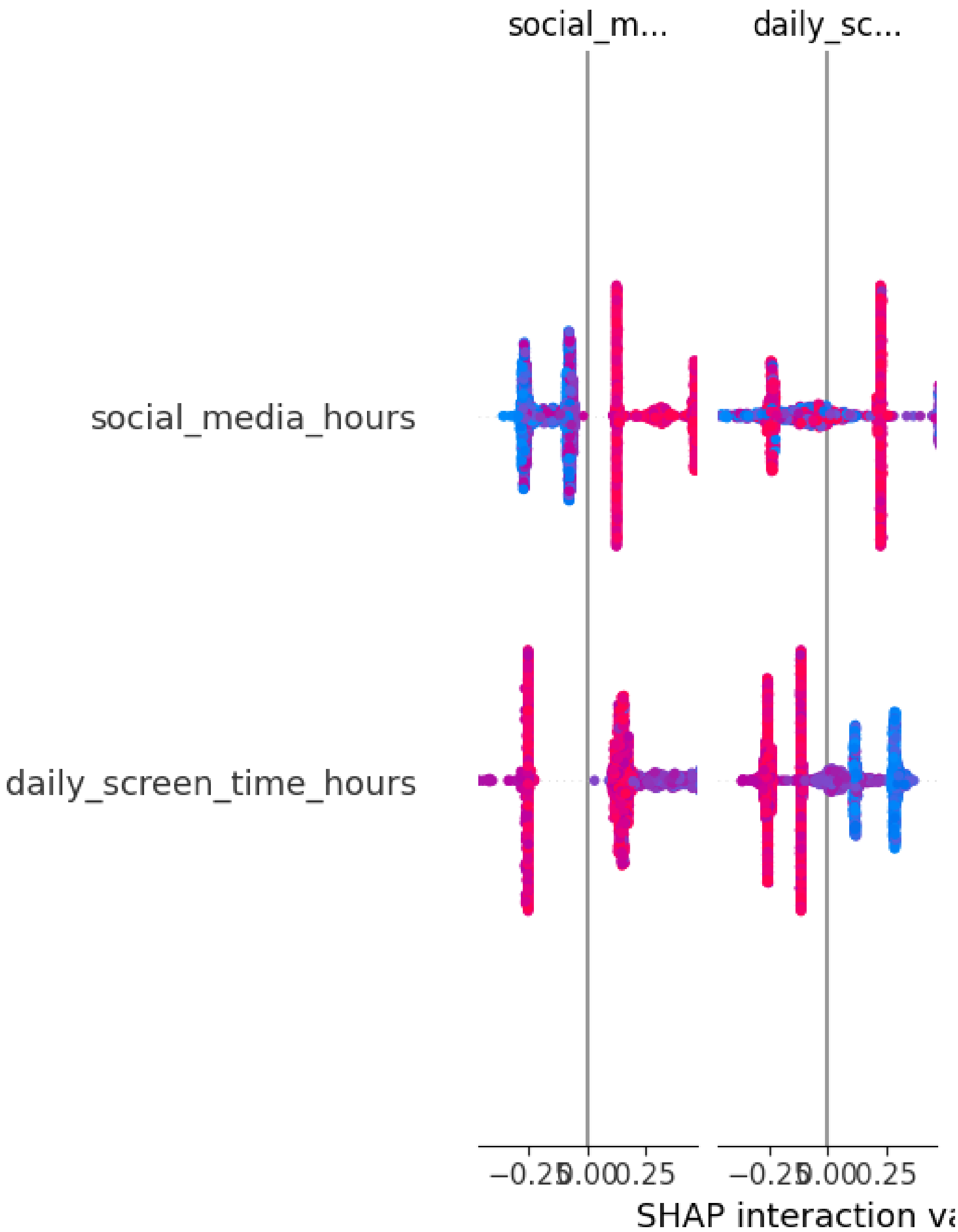

Figure 5. SHAP Explainability Visualization

6. Conclusion and Future Work

6.1 Achievement of Research Objectives

- **Objective 1**
  The first objective was successfully achieved by implementing supervised machine learning models including Logistic Regression and Random Forest for digital burnout prediction.
- **Objective 2**
  The second objective was achieved through evaluating model performance using accuracy, precision, recall, and F1-score metrics. The Random Forest model achieved the highest performance with approximately 94% accuracy.
- **Objective 3**
  The third objective was achieved using SHAP explainability and feature importance analysis, which identified social_media_hours, daily_screen_time_hours, and weekend_screen_time as the most influential burnout predictors.

6.2 Conclusion

This research aimed to predict digital burnout using smartphone usage patterns and explainable machine learning techniques.

The study demonstrated that smartphone behavioral usage patterns can effectively predict digital burnout risk using supervised machine learning classification models.

The Logistic Regression model achieved approximately 89% classification accuracy, while the Random Forest model achieved the highest predictive performance with approximately 94% accuracy.

The findings showed that:

- social_media_hours,
- daily_screen_time_hours,
- and weekend_screen_time

were the most influential behavioral features affecting digital burnout prediction.

The results also confirmed that explainable artificial intelligence techniques such as SHAP improve model interpretability by identifying the behavioral features contributing most to prediction outcomes.

Overall, the research confirmed that smartphone behavioral data is valuable for identifying users who may be at risk of digital burnout and smartphone addiction.

6.3 Future Work

Future work may improve this research in several ways.

First, future studies can use larger and more diverse datasets from different:

- age groups,
- countries,
- occupations,
- and educational backgrounds.

This may improve the generalizability and reliability of machine learning predictions.

Second, future research may include additional behavioral features such as:

- late-night smartphone usage,
- notification response time,
- emotional state,
- physical activity,
- and mental health indicators.

These additional features may improve burnout prediction accuracy and provide deeper behavioral analysis.

Third, deep learning techniques such as:

- Artificial Neural Networks (ANN),
- Long Short-Term Memory (LSTM),
- and Transformer-based models

may be tested and compared with traditional machine learning classification models.

Fourth, a real-time monitoring system or mobile application may be developed to detect early signs of digital burnout and provide personalized recommendations to users.

Finally, future studies may validate machine learning predictions using real participants and compare prediction results with psychological burnout assessment tools and mental health evaluation methods.

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